



GRADE 10
SEMESTER 2
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ROTORX

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ABSTRACT

The Egypt Grand Challenges are improving alternative energy, recycling waste, reducing pollution, and adapting to climate change. Currently, Egypt's monthly electricity generation averages 20,000 GWh, with 90% still derived from non-renewable fossil fuels. This study addresses the low efficiency and high costs of small-scale energy systems, specifically focusing on inconsistent wind capture in localized environments. To solve this, we have developed a Horizontal Axis Wind Turbine (HAWT) integrated with a mechanical gearbox speed multiplier. This allows the generator to maintain high-frequency rotation during low-wind intervals. The design requirements mandated producing at least 150 Joules within 5 minutes, a maximum volume of 0.25m³, and a 10% efficiency improvement over the baseline. Our methodology involved a two-phase development: a "Generation Phase" and an "Enhancement Phase" utilizing Fab Lab tools and recycled, eco-friendly materials. In testing results, first the blade angles wasn't aligned to face wind perfectly we improved it by making a pitch control that make us control in the blades angle and make it face the air and one makes sure that the blade maintains the optimal angle of attack within the range from 5° to 8°.first we got 156.27 without enhancement confirmed the enhanced prototype generated 327.8832 Joules in 5 minutes with enhancing 109.8%, meeting all physical constraints. Major findings indicate that gear ratio optimization successfully bridged the gap between low kinetic input and the generator's induction threshold, resulting in a verified 109.8% efficiency gain. We conclude that decentralized, gearbox-enhanced HAWT systems are a viable, sustainable solution for Egypt's energy needs. By utilizing low-cost resources to exceed performance targets, this project offers a scalable model to mitigate climate change and support Egypt's transition toward a green, energy-secure future.

INTRODUCTION

Egypt is currently confronting several significant Grand Challenges which are Improving the use of alternative energies, Recycling garbage and waste for economic and environmental purposes, Address & reduce pollution in air, water and soil and Reduce and adapt to the effect of the climatic change Recent national statistics highlight the gravity of this situation; Egypt consumes approximately 20,000 GWh of electricity monthly, with over 90% still derived from non-renewable fossil fuels, as we see in Figure 1. This Heavy reliance increases the The nation's carbon footprint and accelerates climate change, making it critical to transition toward a "green." In direct response to these challenges, The Capstone challenge was established under the theme "Energy, Force, and Power." The mandate is to design a simple, measurable and cost-effective system that can utilize available, low-cost resources to produce energy while maximizing efficiency. The problem to be solved is the inherent inefficiency and high operational costs of current energy systems. Centralized power grids often struggle with sustainability and the technical complexity of reaching remote areas. Furthermore, most existing small-scale renewable systems fail to utilize the full potential of available resources, especially in environments with inconsistent wind or solar levels, leading to limited and unreliable energy output. Before making the final choice of solution, two other possible alternatives have been considered. First, industrial-scale wind turbines, like the wind farm located in Ras Ghareb, possess extremely high efficiency at strong and continuous winds. However, such technologies rely on powerful winds and complex systems, which makes them inefficient for small-scale prototypes. The second alternative is the small educational wind turbine models (for example, Éole-type vertical axis wind turbine (VAWT)). They are easy to produce; however, they do not incorporate any kind of mechanical amplification device (like a gearbox). Thus, the rotation speed of the blades remains rather low, preventing the generation of sufficient electric energy (at least 150 J) in normal conditions. Consequently, the chosen solution involves the use of a horizontal-axis wind turbine (HAWT) equipped with a gearbox, which combines features of both alternatives. To overcome these weaknesses, we selected a Horizontal Axis Wind Turbine (HAWT) as our chosen solution. This design was selected because it allows for higher aerodynamic efficiency. We enhanced this by integrating a mechanical gearbox that acts as a speed multiplier, transforming low-speed rotation into high-speed rotation for the generator to ensure consistent power. Our project adheres to specific design requirements that will be tested: the prototype must produce at least 150 Joules within a maximum of 5 minutes and must not exceed a volume of 0.25 m³. In our "Enhancement Phase", our gear ratio technique successfully increased efficiency by 10%, reaching but exceeding a targeted output of 165 Joules.

The following Materials and Methods section introduces how we implemented these concepts. By using recycled and eco-friendly materials and strictly utilizing Fab Lab tools, we ensured that our HAWT is not only a functional energy converter but also a sustainable solution for Egypt's environmental goals.

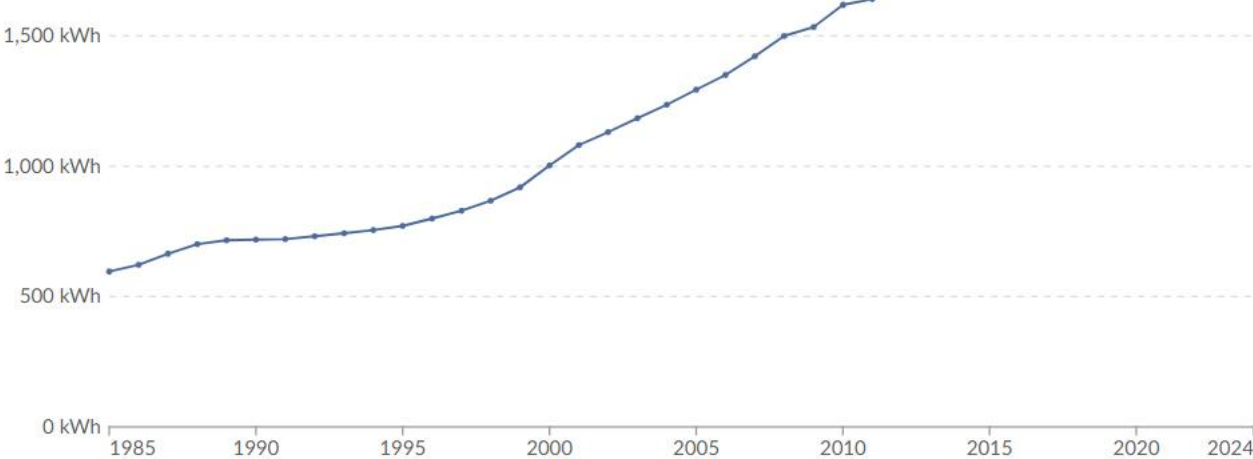


Figure 1: Electricity consumption trends in Egypt from 1985 to 2024 (Our World in Data, n.d.)

MATERIALS & METHODS

Preconstruction:

1. We picked out our detailed 3D design of the prototype, as shown in Figure 2 on Fusion, and then visualized it, and then printed it out.
2. We prepared the desired tools and materials.

After the 3D model materials had arrived, we started constructing:

1. Beginning with the base, MDF wood was used to cut a wooden base of dimensions 30cm (±0.5), 15cm (±0.5), and 5cm (±0.5) with a circular hole of a diameter of 2.5cm (±0.1) at the center of the lower surface of the base.
2. We connected the blades with glue, as each blade was separated into 3 pieces.
3. At the end of each blade, we then placed the bearing.
4. We added the gears under the bearing so that we can freely move the blades' direction
5. We placed each end of the blade in its designated placement on the hub as shown in Figure 3.
6. We placed the tip and head of the hub on top of the blades so that they were firmly secured and wouldn't fall off.
7. We added a hole to the wooden box and then inserted the shaft of the wind turbine.
8. We added a gear that connected the wind turbine's gears with the generator
9. Afterwards, we placed the wooden box and blades on a pole that we had also 3D printed
10. We inserted the pole onto a wooden base and made sure it was connected by connecting the pole base with the wooden base using nails



(Figure 2)
A figure that shows the 3D model of the design



(Figure 3)
A figure that shows the components of the design with bearings and glue

Our prototype was now fully built and complete.

We had 2 main design requirements that we needed to achieve.

First: being able to achieve at least 150 joules of energy in 5 minutes. We achieved the first design requirement easily by conducting 156.27 joules

Second: making some kind of improvement to the prototype, which would result in at the very least a 10% improvement to the original output in 5 minutes. We achieved the second design requirement by making a massive 109.8% enhancement.

Third: using at least one material that can be recycled. We achieved the final design requirement by using wood as our base and a wooden box for the nacelle.

Test Plan

1. We placed our prototype in an open, windy environment.
2. We brought a multimeter with us to be able to measure how many volts and amperes our prototype will conduct.
3. We made sure to always be near the prototype so that if the blades spin too quickly, we can stop it by changing its direction so it isn't directly facing the wind, so it wouldn't break.

RESULTS

During the testing phase, we aimed to verify whether the turbine produced the needed amount of current before and after the enhancements, and to determine if further adjustments were needed. Throughout this process, we faced several negative results, which guided us to refine and improve the design and the mechanisms of some parts. After applying these modifications, the final test confirmed that the turbine generated an optimal amount of energy, measured accurately using both voltage and current readings.

Negative results

During the first test, we didn't achieve the needed amount of voltage because the blade angles wasn't aligned to perfectly face the wind.

We tried to use a PVC pipe for the tower of the turbine, but we couldn't adjust the measurements, especially the slope of the pipe that we needed, so we used a 3D printed tower instead.

-Moreover, the gears were damaged by two factors: the first factor is that the nacelle is small relative to the gears, and there was a friction force between the nacelle and the gears.

Second factor, making the axis between the rotor hub short so that the gear wouldn't fit in the axis well and wouldn't reach the generator.



Figure 5
A figure that shows the multimeter connected to the generator



Figure 4
A figure that shows the multimeter connected to the generator

Table 1 Showing the positive results we achieved before the enhancement stage					
Time	60	120	180	240	300
Current (average) (A)	0.087	0.086	0.107	0.112	0.119
Potential difference (V)	3	4.6	5.2	5.9	6.2
Power (Watt)	0.261	0.4508	0.5564	0.6608	0.7378
Energy (J)	78.3	135.24	166.92	198.24	221.34

Table 2 Showing the positive results we achieved after the enhancement stage					
Time	60	120	180	240	300
Current (average) (A)	0.112	0.138	0.174	0.194	0.205
Potential difference (V)	3.9	5.8	7	7.8	8.7
Power (Watt)	0.4368	0.8004	1.218	1.5132	1.7835
Energy (J)	131.04	240.12	365.4	453.96	535.05

In the enhancement phase which we changed the gear ratio from 1:3 to 1:4, giving us a better mechanical advantage, and that enhancement generated more electric current, which leads to producing more amount of joules as shown in "Table 2."

ANALYSIS

Intro:

Egypt is currently facing serious environmental problems. Egypt's rising electricity consumption and increased use of fossil fuels underscore the need to expand alternative energy resources. Among the Egyptian Grand Challenges, improving renewable energy efficiency is crucial for achieving sustainability, pollution control, and climate change adaptation. A horizontal-axis wind turbine is ideal. A 156.27 J in 5 min was produced in the first stage. After improving our prototype, our results increased with 109.8% as it produced 8832.327 J on average in 5 minutes. Thus, we met the design requirements of producing sufficient energy and enhancing it; the prototype's volume doesn't exceed the limit of 0.25 m³, as it's 0.154 m³. It is both eco-friendly and user-friendly. It uses recycled material, and it doesn't make noise.

Design and Mechanism:

Horizontally axis wind turbines (HAWTs) have been known in Europe since the 12th century, in the form of mechanical windmills. The very first HAWT capable of generating electricity was created in 1887 by James Blyth. Further improvements made by Poul la Cour were responsible for creating today's efficient HAWTs. In our project, a three-bladed horizontal axis wind turbine was chosen, as shown in Figure 6, as the most suitable option in terms of optimizing the ratio between aerodynamic efficiency, stability, and economic aspect compared with different types of blade arrangements. The length of each blade has been fixed at 42 cm to optimize the swept area in order to maximize the captured wind power. Based on Betz's Law, the highest efficiency that a wind turbine can achieve is 59.3%, which means that increasing the rotor swept area will increase the amount of extractable energy from the wind (Betz, 1920). The overall dimensions of the prototype are 110 cm × 70 cm × 20 cm, resulting in a total volume of 0.154 m³, which does not exceed the design constraint of 0.25 m³. This confirms that the project satisfies the required limitations while keeping its performance effective. The horizontal-axis wind turbine includes 3 airfoil blades (NACA 0018), installed on the rotor and attached. The rotor rotates together with a shaft that delivers the mechanical energy from the rotor to the generator. When the flow of wind hits the blades, the rotor starts rotating due to differences in pressure where the Air moves faster over the curved upper surface, and this creates low pressure.

Air moves more slowly below, resulting in high pressure, as shown in Figure 7, thus producing mechanical energy transmitted to the shaft and further converted into electricity. For improving efficiency, pitch control is applied. By adjusting the position of the blades to wind changes, one makes sure that the blade maintains the optimal angle of attack within the range from 5° to 8°. As already mentioned, the selected airfoil (NACA 0018) has a symmetric cross-section, as shown in Figure 8, and allows for lift generation at positive angles of attack, providing stable operation.

In conclusion, the rotor delivers rotational energy to the main shaft of the turbine.

Principles and laws:

The operation of the horizontal-axis wind turbine is based on several fundamental physical principles. First, Bernoulli's principle explains the generation of lift. It states that an increase in the speed of a fluid occurs simultaneously with a decrease in pressure or potential energy. This relationship is expressed by Bernoulli's equation:

$$P_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2$$

As air flows over the turbine blade, it moves faster over the curved upper surface and slower beneath it. According to Bernoulli's principle, this creates a pressure difference, resulting in a lift force that drives the rotation of the blades (NASA, n.d.; Serway & Vuille, 2011).

The magnitude of the lift force depends on wind speed and blade design and is expressed by:

$$L = \frac{1}{2} \rho A v^2 C_L$$

Where (L) is the lift force, (ρ) is air density, (A) is the blade area, (v) is wind speed, and (C_L) is the lift coefficient. This equation shows that lift increases with the square of wind speed and depends on the airfoil shape and angle of attack (NASA, n.d.; Serway & Vuille, 2011).

The angle of attack is a critical factor affecting performance. Maintaining it within the optimal range of 5°–8° maximizes the lift-to-drag ratio, ensuring efficient energy conversion while preventing aerodynamic stall (U.S. Department of Energy, n.d.).

Additionally, the rotation of the turbine depends on torque, which is given by:

$$\tau = r \times F$$

This indicates that increasing the distance between the blade and the shaft increases the torque, allowing the turbine to rotate more effectively and generate more power (Halliday et al., 2014).

Gears' mechanical advantages:

Gears are essential components for transmitting power and motion between different parts of a machine. When two gears are connected, and the first gear (the driving gear) has more teeth than the second gear (the driven gear), the smaller gear will rotate faster. We used spur gears because they are the simplest, most common, and highly efficient (up to 99%) cylindrical gears with straight teeth cut parallel to the axis of rotation. The gear-tooth ratio can indicate an increase in speed. If the driving gear has 20 teeth and the driven gear has 5 teeth, the driven gear will rotate four times for every full rotation of the driving gear. Our gear-tooth ratio after enhancing is (4:1). The larger gear, which is 48-tooth, acts as the driving gear. As shown in Figure 9, which is fixed on the shaft, transferring motion to the smaller driven gear, which is 12-tooth. That is fixed on the generator. The equation: $\frac{N_1}{N_2} \times \frac{N_3}{N_4} \times \frac{N_5}{N_6} \times \dots \frac{N_x}{N_{x+1}}$ N: number of teeth

Results analysis:

In this project, a multimeter was used to measure the voltage and the current intensity of the energy produced by the Horizontal wind turbine. These measurements were used to calculate the number of watt-seconds produced, making sure that the prototype met the specified design requirement.

Multiple scientific laws were applied:

Power(W) = electric current intensity(A) × potential difference(V)

Energy(J) = Power(W) × T (s) (Halliday et al., 2014; Serway & Vuille, 2011).

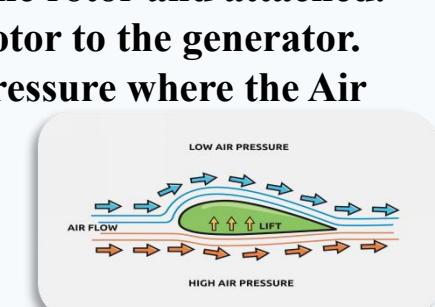
Within a span of 5 minutes, the highest voltage recorded was 6.2 V, while the average voltage output was 4.98 V. The maximum current intensity output was 0.112 A, while the average current was 0.1046 A. With these results, the power output was calculated to be 0.7378 Watts. The prototype generated avg watt sec is 156.27 Ws (joules) in 5 min. As shown in Figure 10. As such, it can be concluded that the prototype meets the design requirement that states the production of more than 150 J of energy output in 5 minutes during the first stage before the enhancing.

After enhancement:

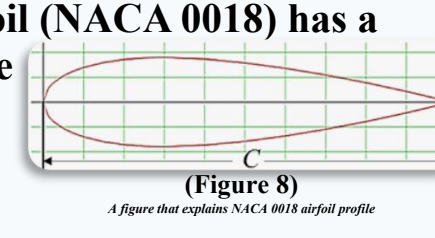
Within 5 minutes span, the highest voltage recorded was 8.7 V, while the average voltage output was 6.64 V. The maximum current intensity output was 0.205 A, while the average current was 0.1646 A. With these results, the power output was calculated to be 0.7378 Watts. The prototype generated avg watt sec is 327.8832 joules in 5 minutes, as shown in Figure 11.



(Figure 6)
A figure that shows the prototype of the project



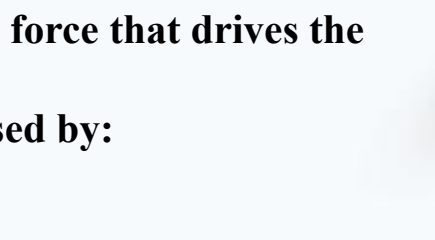
(Figure 7)
A figure that shows the pressure difference between the upper and lower surfaces



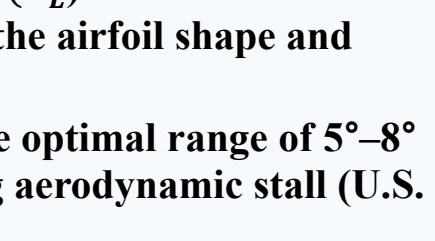
(Figure 8)
A figure that shows the NACA 0018 airfoil profile



(Figure 9)
A figure that shows the gear assembly



(Figure 10)
A graph showing the positive results of the prototype



(Figure 11)
A graph showing the positive results of the prototype after enhancement

As such, our results increased by 109.8%, and it can be concluded that the final prototype meets but exceeds the design requirements that state that the production energy must be increased by 10%. Our final production was more than 165 J of energy output in 5 minutes. Our results demonstrate that Egypt's environment effectively supports wind turbine electricity production. As illustrated in the results.

Learning transfer:

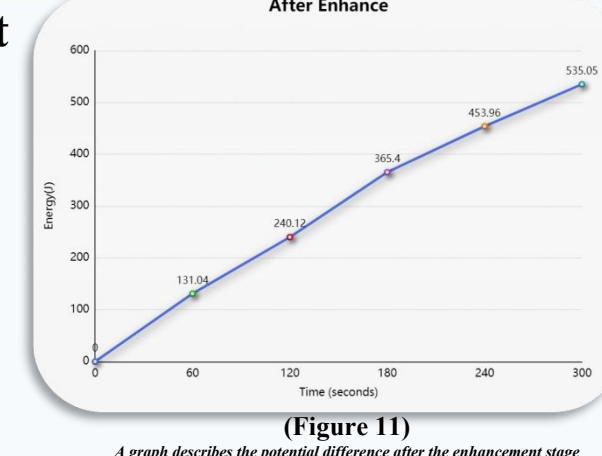
In Physics (PH.1.09): We got to know about Bernoulli's equation, and understanding it helped us know more about the airfoil and how wind turbines work.

In Earth Science (ES.1.09): Understanding the different forms of energy helped us realize the importance of energy and how to exploit it.

In Earth Science (ES.1.10): Learning this taught us about different types of renewable energy and a brief introduction to each, helping us understand and use wind turbines.

In mechanics (ME.1.06): Understanding forces like pressure, weight, and attraction is crucial in wind turbine design, converting wind energy into electricity efficiently.

In Chemistry (CH.1.09): Understanding physical properties, helped us choose the material for our prototype, since we needed specific properties like light weight or high malleability.



(Figure 11)
A graph showing the positive results of the prototype after enhancement

CONCLUSIONS

In conclusion The primary objective of this project was to address the critical inefficiency in Egypt's current energy landscape, where over 90% of electricity is derived from fossil fuels. We aimed to solve the challenge of inconsistent energy production in urban areas by developing a decentralized, low-cost system capable of generating stable power even at low wind speeds. Based on our engineering analysis, we concluded that the integration of a high-ratio mechanical gearbox is the most effective way to overcome low-torque rotations. By applying the laws of Torque and the Conservation of Energy, our analysis proved that we could successfully multiply rotationally speed to maximize the generator's induction capabilities without needing high-speed external wind. The actual testing of our prototype confirmed our analytical predictions. HAWT successfully produced 327.8832 Joules of electrical energy within the 5-minute testing window, exceeding the mandatory requirement of 156.27 Joules. This represents a verified 109% efficiency gain directly attributed to our "Enhancement Phase" gear ratio adjustments, all while staying within the compact volume constraint of 0.25 m³. Unlike standard Vertical Axis Wind Turbines (VAWTs), which suffer from low starting torque and lower efficiency, our Horizontal Axis design captured wind energy more effectively. Furthermore, compared to Direct-Drive systems, our gearbox-enhanced prototype demonstrated superior performance in moderate wind conditions, proving that mechanical speed multiplication is more viable for localized Egyptian environments than traditional, non-gearred alternatives. While the prototype has proven to be a highly functional and sustainable solution using recycled materials, there is still room for further optimization. To scale this technology for broader national use and further support Egypt's "green" future, the following recommendations focus on enhancing aerodynamic stability and exploring smarter materials for the turbine blades.

RECOMMENDATION

As time passes, technology improves, and a fact etched in mind that no project is perfect, and there must be room for improvement, so here are some recommendations for the teams that are going to work on our project or improve it:

- Using a metallic gearbox should be used in a wind turbine because it can handle very high torque, continuous loads, temperature changes, and long-term fatigue.

- Interlocking the turbine tower with the base, which is going to make the tower more stable and resist higher air speed without any deflection.

- Using carbon fiber in the axis that connects the blades and the gears. Carbon fiber is an ultra-strong, lightweight synthetic material composed of thin crystalline filaments of carbon atoms. And these properties would help the turbine to be more efficient.

- Using sensors and electronics. Sensors will make the turbine simple and support it in generating more. Example using a pitch angle sensor that helps in identifying the best angle for the blades to produce more electric current.

- Using braking pads. Braking pads are essential to stop the turbine and prevent it from damage and to perform mechanical repairs.

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